Take-home Final Assignment
Phy 426, 2026
J. Klymak

DUE: 15 Apr, 2026, 14:00

Exam to be completed independently, though questions to the instructor are welcome/encouraged. Open-book, open-notes are fine. Show all work, define any constants you need that I don't provide, check your units, etc. Except as noted, the density of the fluid is ρ, gravity is g, the kinematic viscosity ν, and the fluid can be assumed Bousinesque and incompressible.

Please try to make it readable. I will deduct up to 10% for illegible chicken scratches, so please take the time to recopy your work. Typeset is preferred but not necessary.

Question 1. Shear production of turbulent kinetic energy

Consider a slowly evolving flow in the x-direction with a slowly evolving shear in the z-direction given by $\partial U/\partial z \neq 0$. On top of this slowly evolving flow is a flow that is turbulent with velocity fluctuations u', v', w'

1. [10] Using Reynolds averaging form an energy equation for the mean flow and show that the mean flow has a sink of energy given by $+\overline{u'w'}\frac{\partial U}{\partial z}$
2. [4] Explain why this is a sink instead of a source by considering the likely sign of the terms.
3. [8] Again using Reynolds averaging on the energy equation, show that the *turbulent* kinetic energy has a *source* due to this term.

Question 2. Oscillating forcing of a fluid above a plate

Consider a fluid with viscosity ν that has an infinite plate as a no-slip bottom boundary condition. The fluid is infinite in y above the plate. Suppose that the fluid is everywhere subject to an oscillating horizontal pressure gradient (parallel to the plate), given by

$$\frac{1}{\rho}\frac{\mathrm{d}P}{\mathrm{d}x} = A\cos\omega t \tag{1}$$

where ω is the forcing frequency in rad/s.

1. [15] Show that a solution for the flow as a function of y and t, assuming the flow is well-developed (e.g. long after any startup transient) is given by:

$$u(y,t) = -\frac{A}{\omega}\left(\sin\omega t\left(1 - e^{-ky}\cos ky\right) + \cos\omega t\ e^{-ky}\sin ky\right) \tag{2}$$

   where $k \equiv \sqrt{\frac{\omega}{2\nu}}$.

   Be sure to describe the steps you take to solve this problem, and to clearly state any assumptions you make.

2. [5] Comment on the character of the solution, particularly with respects to the phase of the solution both far from and near the plate. A sketch is helpful - and while you are not required to, I found doing this quickly in a plotting package helpful.

3. [10] Calculate the work per length x done by the pressure force on the flow, and show that it is equal to the viscous dissipation in the flow, averaged over a forcing cycle.

Question 3. Two-layer internal tide generation at a step

Consider a two-layer fluid with a step in the bottom topography. The upper layer has resting depth h_1 and density ρ_1, and the lower layer has a resting depth h_2 and density ρ_2. The density difference is small compared to the absolute density, so $\Delta\rho = \rho_2 - \rho_1 \ll \rho_1, \rho_2$. The bottom topography is $z = -h_1 - h_2$ for $x < 0$ and $z = -h_1 - h_2 + b$ for $x > 0$ if the seasurface is at $z = 0$.

A tide of frequency ω comes from the left ($x < 0$) direction with a seasurface amplitude $\eta_I = A_I \cos(k_{FL}(x - c_{FL}t))$, where $c_{FL} = \sqrt{g(h_1 + h_2)}$ is the *Fast* surface wave speed, and $k_{FL} = \omega/c_{FL}$ is the wavenumber of the incoming tide.

1. [5] Sketch the problem and indicate the waves you expect to see generated by this flow.

2. [10] Solve for the amplitude of the reflected and transmitted surface waves under the assumption that the surface response does not know about the internal waves and argue that

$$A_R \approx \frac{c_{FR} - c_{FL}}{c_{FR} + c_{FL}} A_I \tag{3}$$

and

$$A_T \approx \frac{2c_{FR}}{c_{FR} + c_{FL}} A_I \tag{4}$$

where $c_{FL} = \sqrt{g(h_1 + h_2)}$ and $c_{FR} = \sqrt{g(h_1 + h_2 - b)}$ are the fast surface wave speeds on the left and right sides of the step, respectively. (Hint: you will need two boundary conditions at $x = 0$; one is that the sea surface elevation is continuous, and the other is that the transport of water across the boundary is continuous).

3. [20] At the step ($x = 0$), let the total interface displacement on the left and right be

$$\zeta^-(0) = \zeta_I^s + \zeta_R^s + \zeta_L^i \tag{5}$$

and

$$\zeta^+(0) = \zeta_T^s + \zeta_R^i \tag{6}$$

where ζ_I^s, ζ_R^s, and ζ_T^s are the interface displacements induced by incoming, reflected, and transmitted *surface* waves, and ζ_L^i and ζ_R^i are the generated *internal* waves radiating to the left and right, respectively.

The first boundary condition is continuity of interface displacement:

$$\zeta^-(0) = \zeta^+(0) \tag{7}$$

with $\zeta_I^s = \alpha_L A_I$, $\zeta_R^s = \alpha_L A_R$, and $\zeta_T^s = \alpha_R A_T$.

The second boundary condition is continuity of lower-layer transport at $x = 0$:

$$Q_2^-(0) = Q_2^+(0) \tag{8}$$

with

$$Q_2^-(0) = h_2\left(u_{2I} - u_{2R} + u_{2L}^i\right), \qquad Q_2^+(0) = (h_2 - b)\left(u_{2T} + u_{2R}^i\right) \tag{9}$$

where the minus sign on u_{2R} accounts for leftward propagation of the reflected wave.

Use these two conditions to solve for

$$\zeta_L^i = \frac{S + c_{IR}\Delta}{c_{IL} + c_{IR}} \tag{10}$$

and

$$\zeta_R^i = \frac{S - c_{IL}\Delta}{c_{IL} + c_{IR}} \tag{11}$$

where

$$\Delta = \alpha_R A_T - \alpha_L A_I - \alpha_L A_R \tag{12}$$

and

$$S = \frac{h_2}{h_1 + h_2}\left(A_I - A_R\right)c_{FL} - \frac{h_2 - b}{h_1 + h_2 - b}A_T c_{FR} \tag{13}$$

and $\alpha_L = \frac{h_2}{h_1+h_2}$, $\alpha_R = \frac{h_2-b}{h_1+h_2-b}$ are geometric parameters, $c_{IL} = \sqrt{g'\frac{h_1 h_2}{h_1+h_2}}$, and $c_{IR} = \sqrt{g'\frac{(h_1)(h_2-b)}{h_1+h_2-b}}$ are the internal wave speeds on the left and right sides of the step, respectively.

4. [10] So lets check if this all makes sense: Suppose $\Delta\rho = 1\text{kg/m}^3$, $h_1 = 20\text{m}$, $h_2 = 100\text{m}$, $b = 50\text{m}$, and $A_I = 1\text{m}$. Calculate the amplitudes of the reflected and transmitted surface waves, and the internal waves radiating to the left and right. *Comment* on the relative sizes of these waves compared to the surface forcing.

5. [15] Given the amplitudes of the internal waves, calculate the energy flux of the internal waves radiating to the left and right, and compare it to the energy flux of the incoming surface tide. *Comment* on the *efficiency* of internal tide generation at this step (eg what fraction of surface energy goes into internal tidal energy) (Hint: the energy flux of a wave is given by Ec_g where E is the energy density of the wave and c_g is the group velocity of the wave. The energy density of a surface wave is given by $E = \frac{1}{2}\rho g A^2$, and the energy density of an internal wave is given by $E = \frac{1}{2}g'\zeta^2$, and recall there is one radiating in each direction)